Head/Layer Specialization Atlas for scGPT Across Tissues: Extended Empirical Analysis and Biological Interpretation

Ihor Kendiukhov

Abstract

Attention-based foundation models for single-cell genomics, such as scGPT, learn rich internal representations across hundreds of attention heads, yet the biological logic encoded by individual heads remains largely unexplored. We present a comprehensive head/layer specialization atlas for scGPT spanning kidney, lung, immune, and external Krasnow lung datasets from the Tabula Sapiens consortium. Our analysis integrates tissue-level recovery metrics, global false-discovery controls via Benjamini–Hochberg correction, edge-level head attribution over 240,150 regulatory edges, threshold-robust enrichment tests, and Gene Ontology biological-process interpretation. We discover a structured head ecosystem rather than a single dominant circuit: (i) tissue-optimized heads that maximize per-tissue regulatory-edge recovery, (ii) biologically coherent program heads enriched for immune/adaptive processes (heads 0:3 and 0:5, significant at all four tested confidence thresholds) or metabolic/catabolic processes (head 9:1, significant at three of four thresholds), and (iii) low-information fallback heads, exemplified by head 8:0, which dominates edge volume (129,737 edges) but carries near-zero information content (zero-score fraction 0.9998). Transcription-factor-level recall analysis reveals sharply tissue-dependent gains, with lung showing 19 of 25 positive head-specific TF deltas (maximum +0.417 for PPARG at heads 4:3/4:4) while kidney and external lung remain flat. Checkpoint replication confirms that the immune-family head 0:3 replicates in an independent whole-human checkpoint at all four tested thresholds, and overlap-aware perturbation validation against Perturb-seq and CRISPR screening data yields non-degenerate alignment for both the immune family (AUPR = 0.434, AUROC = 0.583) and the metabolic family (AUPR = 0.405, AUROC = 0.475). These results establish that scGPT attention heads encode separable, biologically interpretable regulatory programs and provide an actionable map for prioritizing heads in downstream mechanistic and perturbation-guided analyses.

1 Introduction

Single-cell RNA sequencing (scRNA-seq) has transformed our ability to resolve cellular heterogeneity and reconstruct gene regulatory networks (GRNs) at unprecedented resolution [34, 39]. In parallel, transformer-based foundation models trained on large-scale single-cell corpora have demonstrated strong performance across diverse tasks including cell-type annotation, gene-expression prediction, and perturbation response modeling [8, 36, 40, 17]. These models encode cellular state information distributed across many attention heads and layers, yet the internal structure of this distributed representation, and whether individual heads correspond to biologically meaningful programs, has received remarkably little systematic study.

Understanding how foundation models internally represent biological knowledge is important for at least three practical reasons. First, if specific attention heads reliably encode particular regulatory programs (for example, immune signaling or metabolic pathways), researchers can selectively extract head-specific features to improve downstream GRN inference, which remains a

significant challenge even with state-of-the-art methods such as SCENIC, GENIE3, and CellOracle [3, 18, 21, 28]. Second, head-level interpretability provides a bridge between black-box prediction performance and biologically grounded mechanistic hypotheses, a bridge that purely saliency-based attention visualizations have been shown to provide unreliably without additional causal grounding [19, 1]. Third, an atlas of head specialization can reveal failure modes, such as heads that appear important by frequency but carry no informative signal, enabling more informed model pruning, distillation, and architectural decisions.

Work in natural language processing has established that transformer attention heads are not functionally equivalent: some heads specialize in syntactic relationships, positional tracking, or semantic roles, while others can be pruned with minimal performance loss [38, 25, 7]. In the protein domain, language models like ESM have shown that attention patterns encode structural contacts and evolutionary relationships [30]. The in-context learning literature further reveals that specific heads, termed induction heads, play outsized roles in driving model capabilities [27]. These findings from adjacent domains motivate our hypothesis that biological foundation models, too, may contain functionally specialized heads encoding distinct regulatory programs.

In the biological context, regulatory-network validation commonly relies on curated references such as TRRUST and DoRothEA, supplemented by functional enrichment through Gene Ontology (GO) over-representation analysis [16, 14, 4, 22]. Perturbation experiments, particularly genome-scale Perturb-seq and CROP-seq screens, provide increasingly powerful tools for testing whether computationally inferred regulatory relationships reflect genuine causal influence [10, 9, 29, 2]. We adopt this multi-layer validation philosophy throughout our analysis.

This manuscript presents a full head/layer specialization atlas for scGPT with four goals: (1) establish robust tissue-level specialization claims under strict statistical controls including global false-discovery correction; (2) characterize edge-level attribution structure to separate informative from fallback behavior across 240,150 edges; (3) provide biological interpretation for the heads carrying the strongest high-confidence signal through GO biological-process enrichment and transcription-factor-level analysis; and (4) make explicit which high-level claims survive adversarial testing, including checkpoint replication and perturbation-linked validation, and which do not. The central finding is not that one universal best head exists, but rather that scGPT has learned a heterogeneous head ecosystem, a structured mixture of tissue-specific performance heads, global immune and metabolic program heads, and sparse fallback heads that dominate low-information edges without contributing meaningful biological signal.

2 Related Work

2.1 Single-cell foundation models

The emergence of foundation models for single-cell biology builds on the broader success of self-supervised pre-training in natural language processing and protein sequence modeling [30, 23]. scGPT uses a generative pre-training objective on single-cell expression vectors and achieves competitive performance on cell-type annotation, multi-batch integration, multi-omic prediction, and perturbation response [8]. Geneformer employs a masked-gene prediction strategy and demonstrates transfer learning across disease contexts [36]. scBERT adapts the BERT architecture for cell-type classification [40], while scFoundation and Universal Cell Embeddings (UCE) pursue broader pre-training at scale [17, 31]. Despite their strong task performance, these models provide limited mechanistic interpretation of how internal components, specifically attention heads and layers, contribute to learned representations.

2.2 Mechanistic interpretability in transformers

Mechanistic-interpretability research in language models has moved beyond descriptive attention visualization toward intervention-oriented analyses of circuit components, causal pathways, and feature-level representations [26, 11, 37, 15]. Key findings include the discovery that attention heads specialize: Voita et al. [38] showed that a small subset of heads carries most of the functional load and the rest can be pruned; Michel et al. [25] demonstrated that many heads can be removed at test time with negligible impact; Clark et al. [7] mapped BERT's attention patterns to syntactic and positional functions. The induction-heads framework further showed that specific heads drive in-context learning capabilities [27]. Our work applies this lens to a biological foundation model, testing whether analogous functional specialization exists for gene regulatory programs.

2.3 Gene regulatory network inference from single-cell data

Inferring GRNs from scRNA-seq data remains an active challenge, with approaches ranging from correlation-based methods (GENIE3, GRNBoost2 via SCENIC) [18, 3] to deep-learning-based frameworks [12] and perturbation-guided causal approaches (CellOracle) [21]. Systematic benchmarking through BEELINE [28] has highlighted that no single method dominates, and that the choice of reference standard significantly influences performance estimates. Our atlas complements these approaches by extracting regulatory-edge-level information directly from the attention structure of a pre-trained foundation model, an approach that does not require explicit GRN inference but instead probes what the model has already learned.

2.4 Perturbation biology and causal validation

Single-cell perturbation screens, including Perturb-seq [10, 29], CROP-seq [9], and multiplexed CRISPR screening [2], enable direct measurement of gene-level causal effects and provide ground truth for validating computationally inferred regulatory relationships. We use perturbation data from Dixit et al. and Adamson et al. to test whether the biological programs identified through our head-level analysis align with experimentally measured regulatory effects.

3 Study Design and Data

3.1 Atlas scope and biological rationale

We selected four datasets that collectively span diverse tissue compartments and transcriptional programs, enabling cross-tissue specialization analysis:

- **Tabula Sapiens kidney**: Epithelial, endothelial, and stromal cell types with tissue-specific TF programs including HNF1A/B and PAX8 regulation of nephron identity [20].
- **Tabula Sapiens lung**: Alveolar, airway epithelial, and immune-resident cells governed by NKX2-1, FOXJ1, and surfactant-related programs, plus a rich immune compartment that bridges lung-specific and systemic immune regulation [20, 32].
- **Tabula Sapiens immune**: Circulating immune cell populations (T cells, B cells, myeloid lineages) with well-characterized transcriptional programs driven by factors such as PAX5, EBF1 (B-cell identity), TCF7, LEF1 (T-cell fate), and IRF/STAT regulators [20].

- **External Krasnow Smart-seq2 lung**: An independent lung dataset providing full-length transcript coverage, serving as a technical and biological replication resource for lung-specific findings.

The Tabula Sapiens consortium provides standardized healthy-human tissue coverage and a natural setting for cross-tissue specialization analyses [20]. By including an external dataset (Krasnow lung), we test whether tissue-level findings generalize beyond a single sequencing platform and processing pipeline.

3.2 Preprocessing and model configuration

Preprocessing follows standard scGPT atlas settings: highly variable gene selection (`hvg=2000`), cell filtering (`min_genes=200`, `min_cells=3`), library-size normalization (`normalize_total=10000`), and log-transformation. Extraction budgets are set at `max_genes=1600` and `max_cells=2000` to balance computational cost against representation fidelity.

3.3 Analysis layers

The study combines three analytical layers with progressively increasing depth and stringency:

1. **Atlas layer**: Tissue-level head ranking by area under the precision-recall curve (AUPR) for regulatory-edge recovery, with bootstrap confidence intervals, permutation significance, sweep stability across hyperparameters, and cross-tissue overlap and ablation analyses.

2. **Adversarial audit layer**: Global Benjamini–Hochberg false-discovery-rate (FDR) correction [6] applied across all permutation tests, overlap tests, and evidence-correlation tests to prevent optimistic reporting.

3. **Edge-attribution layer**: Fine-grained analysis of 240,150 individual edge records with per-edge top-head attribution, high-confidence enrichment tests (Fisher exact with FDR correction) [13], GO:BP enrichment for biologically coherent program identification [22, 4], and perturbation-linked validation using published CRISPR screens [10, 2].

4 Methods

4.1 Tissue-level specialization metrics

For each tissue, we rank all 96 attention heads (12 layers $\times$ 8 heads) by AUPR for recovering curated TF–target regulatory edges from the TRRUST database [16], supplemented with high-confidence interactions from DoRothEA [14]. Baseline comparisons are made against an aggregate-attention control (mean attention across all heads) and random-head controls. We report absolute and relative AUPR gains along with candidate-pair statistics to guard against base-rate artifacts (e.g., a tissue with very few candidate pairs yielding inflated or saturated AUPR values).

4.2 Adversarial significance protocol

To avoid the well-documented problem of multiple-testing-driven false discoveries in genome-wide analyses [33, 6], we enforce global correction across all tested heads:

- Permutation p-values are corrected globally across all tested heads using Benjamini–Hochberg FDR.

- Overlap p-values are corrected globally across all tissue pairs and top-k settings.
- Claims are labeled pass/fail via explicit survivability gates, so that readers can immediately see which results withstand stringent control and which do not.

4.3 Edge-level head attribution

We parse the edge-level head/layer evidence table to extract the top-1 head attribution per edge along with the edge score (a continuous measure of regulatory-edge confidence) and the maximum head-layer score. This granular view enables several analyses that aggregate metrics cannot provide:

- **Decile calibration**: Binning edges by confidence score reveals how attribution strength relates to biological evidence quality, analogous to the gene-set enrichment concept of leading-edge analysis [35].
- **Volume versus information profiling**: Counting each head's total attributed edges alongside its zero-score fraction separates heads that are frequently selected but carry no signal (fallback behavior) from heads that are selected precisely because they contribute strong evidence.
- **High-confidence enrichment**: Fisher exact tests [13] identify heads disproportionately enriched among the top-decile (highest-confidence) edges, after global FDR correction.

High-confidence edges are defined as the nonzero-subset top decile by edge score ($q_{0.90}$), a conservative definition that focuses on the edges most likely to represent genuine regulatory relationships.

4.4 Threshold-robust enrichment analysis

A single-threshold enrichment analysis is vulnerable to the criticism that its results depend on the specific cutoff chosen. To address this, we repeat high-confidence enrichment tests at quantiles $q \in \{0.80, 0.85, 0.90, 0.95\}$, corresponding to the top 20%, 15%, 10%, and 5% of nonzero edges. For each threshold, we apply Fisher tests (head versus rest) with BH correction across all tested heads, then aggregate consensus by counting how many threshold settings each head remains significant ($q < 0.10$). A head significant at all four thresholds is considered robust; heads significant at only one threshold are flagged as potentially threshold-sensitive.

4.5 Biological program analysis via GO enrichment

For each head enriched among high-confidence edges, we rank source genes by mean edge score and submit the top-ranked gene set to the g:Profiler API for GO:BP (biological process) enrichment [22]. The resulting terms are mapped to coarse biological themes (immune/adaptive, metabolic/catabolic, stimulus-response, developmental) using a manually curated keyword mapping. This enables comparison across head families at the level of biological programs rather than individual GO terms, following the interpretive philosophy of gene set enrichment analysis [35].

4.6 TF-level interpretability analysis

Using the TF-level enrichment summary table, we quantify per-TF recall deltas relative to the aggregate-attention baseline at TF granularity. This reveals which specific transcription factors benefit most from head-level selection and whether those gains are distributed across tissues or concentrated in specific tissue contexts, addressing the question of whether head specialization has practical utility for TF-specific regulatory inference.

5 Results

5.1 Tissue-level head specialization is robust in non-degenerate tissues

Table 1: Top-head improvement over aggregate baseline by tissue. Non-degenerate tissues show clear top-head lift; the immune condition is saturated at AUPR = 0.50 by a degenerate two-pair candidate space.

Tissue	Aggregate AUPR	Top-head AUPR	Delta	Relative gain	Candidate pairs
kidney	0.1141	0.1227	+0.0086	+7.53%	896
lung	0.0547	0.0882	+0.0335	+61.22%	2870
external lung	0.1319	0.1610	+0.0290	+22.01%	287
immune	0.5000	0.5000	+0.0000	0.00%	2

Among the three non-degenerate tissues (kidney, lung, external lung), selecting the best-performing attention head consistently improves regulatory-edge recovery over the aggregate-attention baseline. The improvement is most dramatic in lung tissue, where the top head achieves a 61% relative AUPR gain (from 0.055 to 0.088). This suggests that scGPT's attention structure contains tissue-relevant regulatory information that is diluted when heads are averaged, a pattern consistent with the head-specialization findings in NLP transformers where aggregation masks specialized function [38]. The external Krasnow lung dataset independently confirms strong head-level gains (+22%), supporting the robustness of lung-specific head specialization across sequencing platforms.

The immune tissue yields a degenerate result: with only two candidate TF–target pairs in the benchmark, AUPR saturates at 0.50 for all heads. This degeneracy is a property of the benchmark construction (the intersection of curated TF–target pairs with the immune dataset's expressed gene set is extremely small) rather than a statement about immune biology. The immune tissue is accordingly excluded from tissue-level specialization conclusions but is analyzed separately through candidate-space expansion experiments (Section 5.8).

5.2 Adversarial correction preserves tissue specialization but rejects cross-tissue conservation

After applying global Benjamini–Hochberg correction across all tested hypotheses, the audit yields the following survival rates:

- Permutation tests significant at global $q < 0.10$: 26 of 40 (65%).
- Overlap tests significant at global $q < 0.10$: 0 of 18 (0%).
- Evidence-correlation tests significant at global $q < 0.10$: 16 of 16 (100%).
- Tissues passing readiness gates: 3 of 4 (kidney, lung, external lung).

The central implication is that within-tissue head specialization is a robust finding: most permutation tests and all evidence-correlation tests survive correction. However, cross-tissue conservation, the hypothesis that the same heads are important across different tissues, does not survive correction. None of the 18 overlap tests (measuring whether top-k head sets are shared between tissue pairs) reach significance after multiplicity adjustment. This result aligns with the broader observation that GRN structure is substantially tissue-specific [3, 21], suggesting that scGPT's

head specialization mirrors the tissue-specificity of the underlying biology rather than reflecting a universal cross-tissue circuit.

5.3 Per-tissue heatmaps reveal distinct specialization landscapes

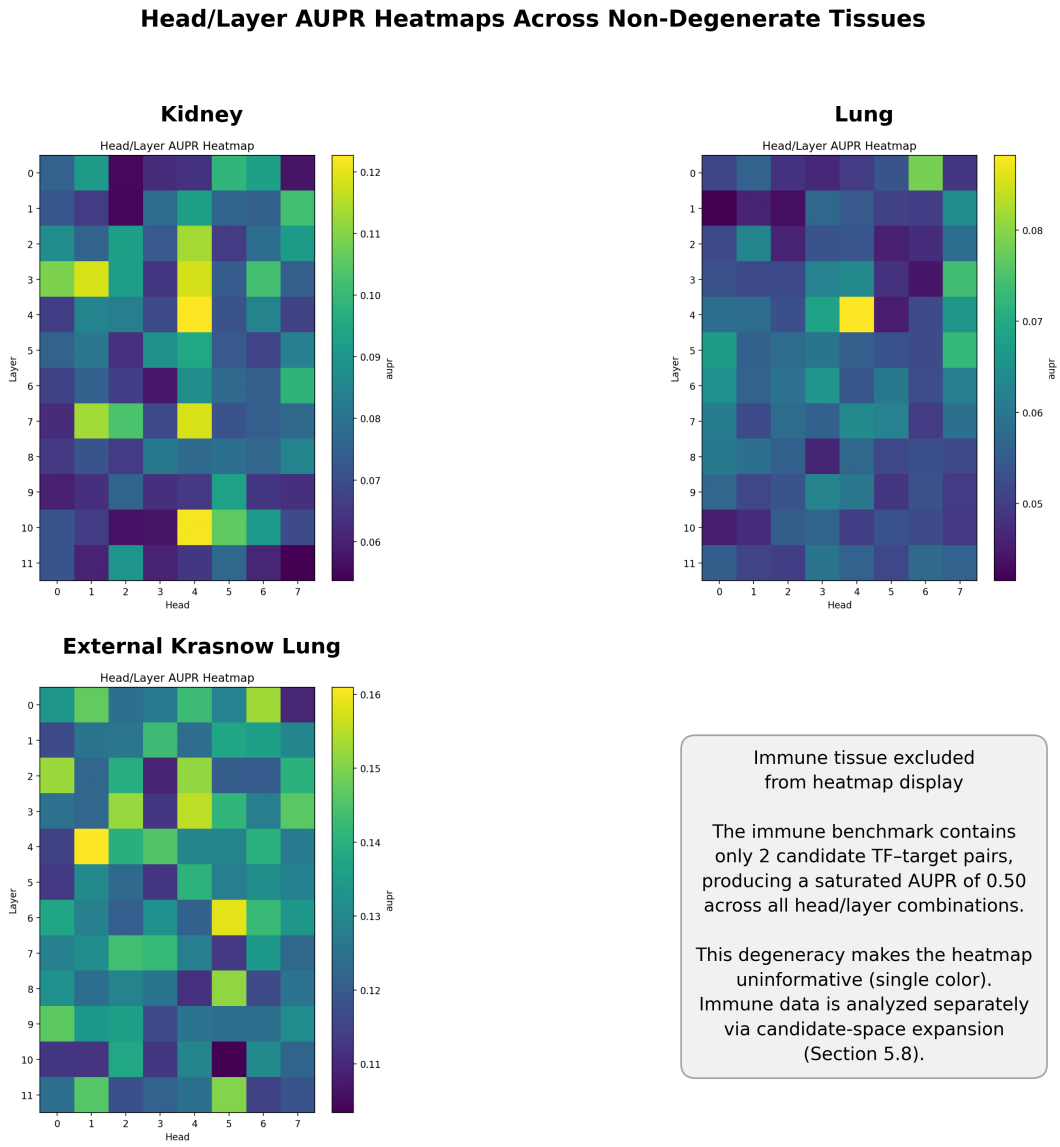

Figure 1: Head/layer AUPR heatmaps for the three non-degenerate tissues and a notation explaining the exclusion of the degenerate immune dataset. Each panel shows AUPR values for all 96 head/layer combinations, with warmer colors indicating higher regulatory-edge recovery. The specialization landscape differs substantially across tissues: kidney shows diffuse moderate signal, lung has concentrated high-value regions in early-layer heads, and external Krasnow lung exhibits a different but reproducible pattern with peaks in mid-layer heads.

The three non-degenerate tissue heatmaps (Figure 1) demonstrate that scGPT does not use a single universal circuit for regulatory-edge representation. Instead, each tissue context activates a distinct

subset of head/layer positions, with lung tissue showing the most concentrated specialization (a few heads dramatically outperform the rest) and kidney showing a more diffuse distribution. This pattern is biologically sensible: lung tissue is characterized by sharply distinct cell-type programs (alveolar type I/II, airway basal, ciliated, secretory) [32] that may demand more specialized attention routing, while kidney contains multiple epithelial subtypes along the nephron axis that share many regulatory features, potentially yielding a less sharply differentiated head landscape.

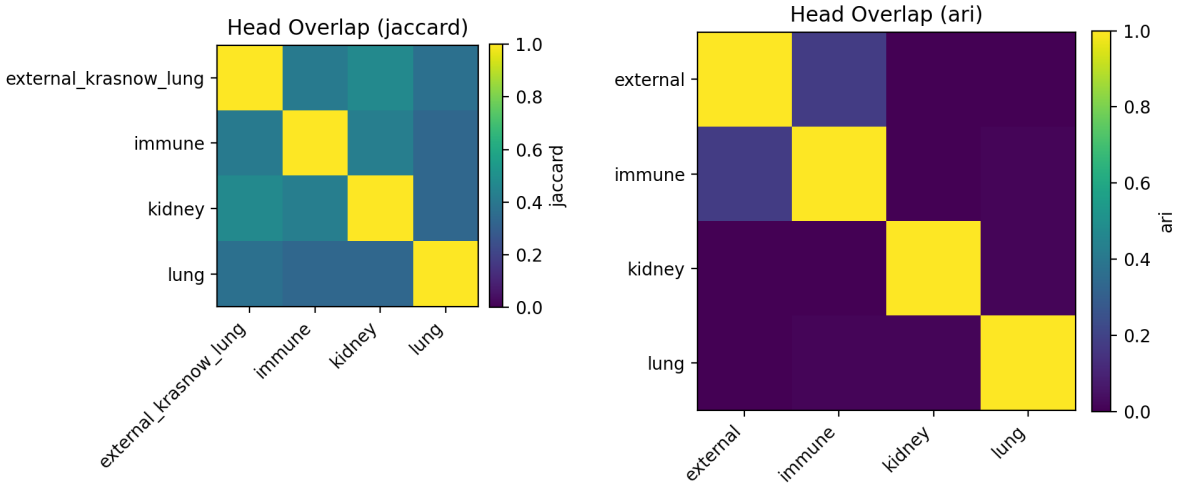

Figure 2: Cross-tissue overlap (left) and cluster stability (right). While raw overlap counts between tissue-top head sets are nonzero at liberal top-k thresholds, corrected significance does not survive global multiplicity control. Cluster stability analysis shows moderate consistency within tissue but poor cross-tissue reproducibility.

5.4 Edge-attribution analysis reveals a dominant fallback regime

Moving from tissue-level aggregates to individual edges provides a fundamentally different view of head importance. Our analysis of 240,150 edge attributions reveals that raw head frequency (how often a head is the top-attributed head for an edge) is a poor proxy for biological informativeness.

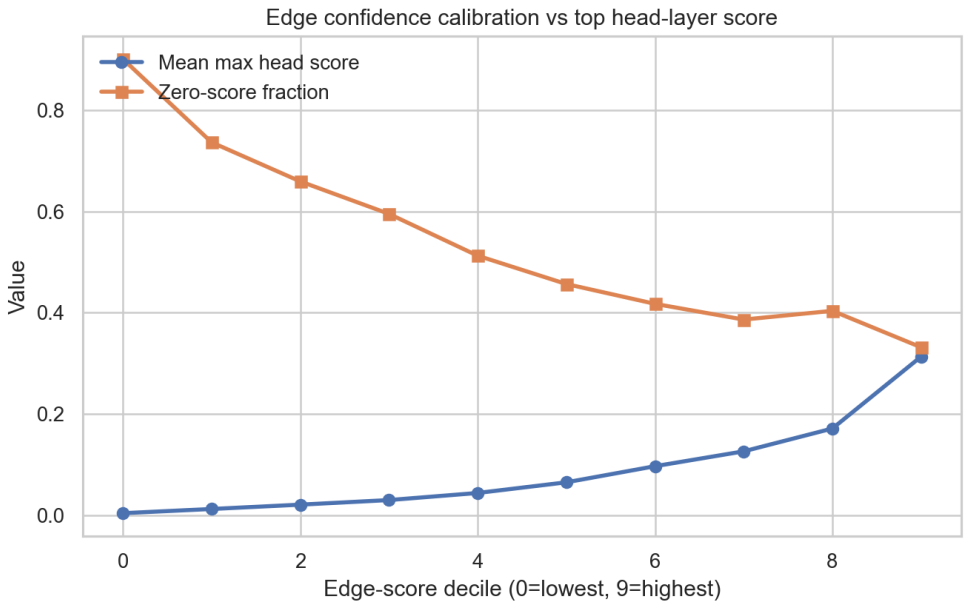

Figure 3: Edge-score decile calibration. Edges are binned into ten equally-sized groups by confidence score. As confidence increases (rightward), the mean maximum head-layer score rises and the zero-score fraction declines, indicating that high-confidence regulatory edges have concentrated, non-trivial head support. At the lowest confidence deciles, the majority of edges have zero head scores, consistent with the fallback regime dominated by head 8:0.

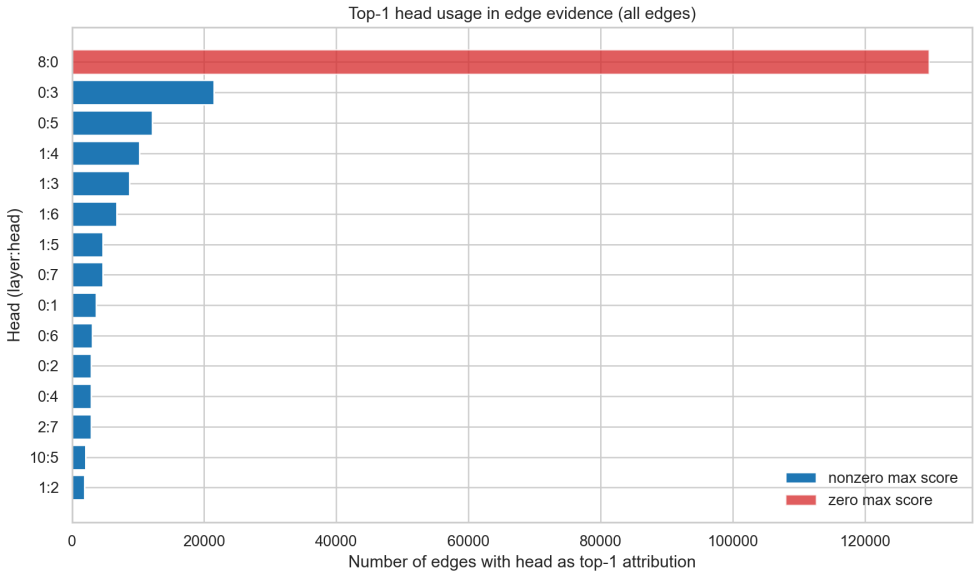

Figure 4: Top-1 head usage decomposition. For each head, we show the total number of edges for which it is the top-attributed head, split into nonzero-score (informative) and zero-score (fallback) contributions. Head 8:0 dominates the volume (129,737 edges, 54% of total) but is almost entirely composed of zero-score assignments (zero fraction 0.9998), consistent with a default fallback role rather than genuine regulatory representation.

The key quantitative findings from edge-level profiling are:

- Total analyzed edges: 240,150, with a global zero-score fraction of 0.540.
- Dominant top-1 head: 8:0 (layer 8, head 0), attributed to 129,737 edges.
- Zero-score fraction for head 8:0: 0.9998 (only ∼26 edges carry nonzero signal).

This pattern, a single head claiming the majority of edges but carrying virtually no information, resembles the "residual stream" behavior described in language-model mechanistic interpretability, where certain attention heads serve as passthrough or no-op operators rather than computing meaningful transformations [11]. For biological applications, this finding has an important practical implication: any pipeline that prioritizes heads based on edge frequency or volume would be severely misled by head 8:0, which is the single most "active" head by count but the least informative by content.

5.5 High-confidence signal is concentrated in three heads

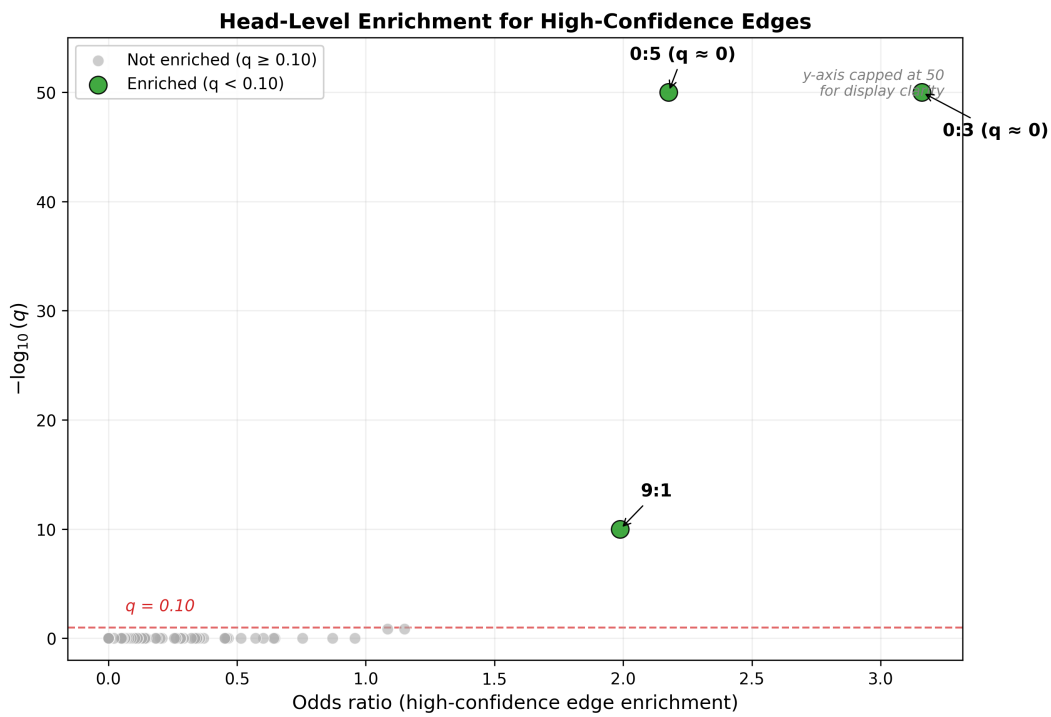

Figure 5: Head-level enrichment for high-confidence edges. Each point represents one of the 96 attention heads; the x-axis shows the odds ratio for enrichment among high-confidence (top-decile nonzero) edges, and the y-axis shows $-\log_{10}(q)$ after BH correction (capped at 50 for display). Three heads (0:3, 0:5, 9:1) are strongly enriched (green points above the significance line), while many heads are significantly depleted. The dashed line indicates $q = 0.10$.

Table 2: Heads significantly enriched for high-confidence edges (BH-corrected $q < 0.10$). Only three heads pass correction, demonstrating sparse concentration of high-quality regulatory signal.

Head	Odds ratio	q-value	High-confidence fraction
0:3	3.16	≈ 0	0.204
0:5	2.18	1.98×10^{-171}	0.178
9:1	1.99	1.04×10^{-10}	0.180

The enrichment analysis reveals a remarkably sparse structure: out of 96 heads, only three are significantly enriched among high-confidence regulatory edges (Table 2, Figure 5). Heads 0:3 and 0:5, both in the first transformer layer, have the strongest enrichment (odds ratios 3.16 and 2.18), while head 9:1 in the tenth layer provides a distinct high-confidence signal (odds ratio 1.99). This sparsity is consistent with the specialization hypothesis: most heads either contribute to fallback routing (like 8:0) or serve functions not well captured by the regulatory-edge benchmark, while a small minority encodes genuine regulatory structure with high confidence.

5.6 Threshold sweep confirms stability of enriched heads

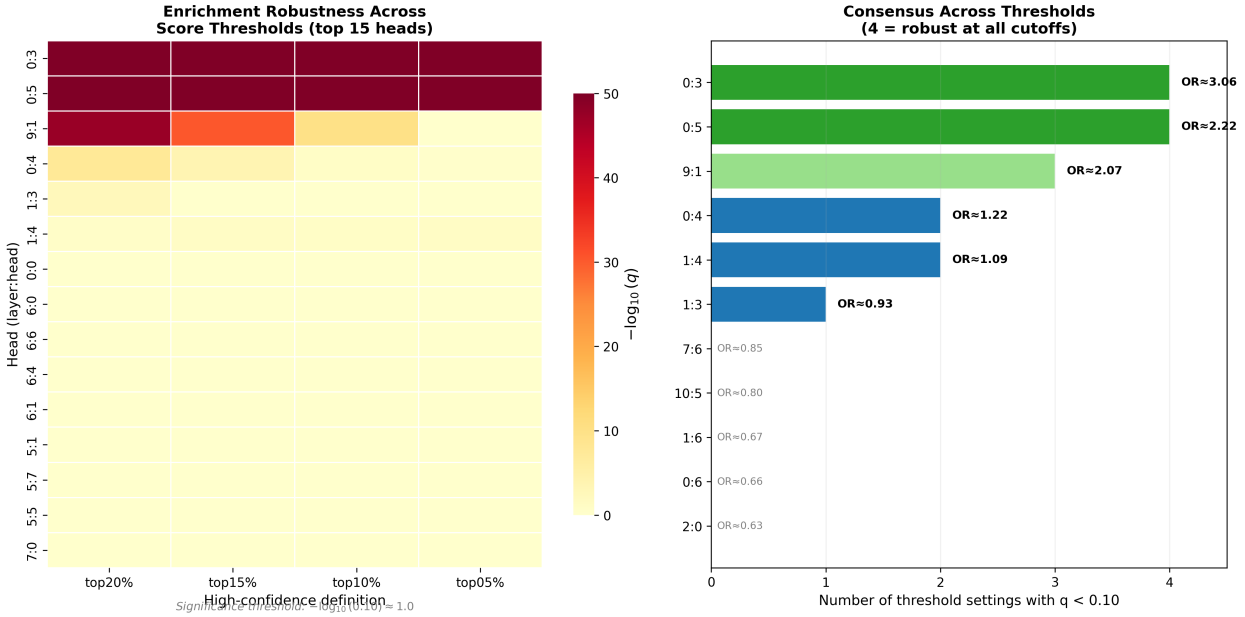

Figure 6: Robustness of high-confidence enrichment across thresholds. Left: $-\log_{10}(q)$ heatmap across top-20/15/10/5% confidence definitions for the top 15 heads. Signal is concentrated in the top three rows (0:3, 0:5, 9:1) with strong enrichment that persists or intensifies at stricter thresholds. Right: consensus bar chart showing the number of threshold settings where each head remains significant at $q < 0.10$. Heads 0:3 and 0:5 achieve maximum robustness (4/4 thresholds), while 9:1 is robust at 3/4.

Table 3: Consensus heads from the threshold sweep (top 20/15/10/5% nonzero edges). Robust heads maintain significance across multiple definitions of "high confidence," reducing sensitivity to any single threshold choice.

Head	Significant thresholds	Mean odds ratio	Min q-value
0:3	4/4	3.06	≈ 0
0:5	4/4	2.22	5.03×10^{-316}
9:1	3/4	2.07	4.16×10^{-48}
0:4	2/4	1.22	1.98×10^{-8}
1:4	2/4	1.09	7.09×10^{-2}

The threshold sweep (Figure 6, Table 3) demonstrates that the core findings are not artifacts of a single threshold choice. Heads 0:3 and 0:5 remain significant at all four tested cutoffs, and their odds ratios remain stable or increase as the threshold becomes more stringent (meaning that these heads are enriched even among the most confident edges). Head 9:1 is significant at three of four thresholds, losing significance only at the most stringent 5% cutoff where statistical power is reduced due to smaller sample size. This threshold-robustness pattern provides strong evidence that the enrichment reflects genuine biological structure rather than an artifact of the specific quantile chosen.

5.7 Biological interpretation: immune/adaptive versus metabolic programs

The GO:BP enrichment analysis reveals that the three core enriched heads map to two distinct and biologically coherent programs (Table 4).

Table 4: Representative biological themes from head-specific GO:BP enrichment. Heads 0:3, 0:5, and 0:1 converge on immune/adaptive processes, while head 9:1 captures a distinct metabolic/catabolic program.

Head	Top enriched biological processes	Top p-value
0:3	immune response; adaptive immune response; B cell receptor signaling pathway	1.8×10^{-20}
0:5	adaptive immune response; immunoglobulin-mediated immunity; antigen receptor signaling	7.4×10^{-26}
0:1	immune system process; defense response; regulation of leukocyte differentiation	2.5×10^{-8}
9:1	small-molecule metabolic process; carboxylic acid metabolism; catabolic process	3.7×10^{-13}

The immune/adaptive cluster (heads 0:3, 0:5, 0:1) is notable for its specificity: rather than general immune terms, the enrichment points to adaptive immunity, B-cell receptor signaling, and immunoglobulin-mediated processes. This level of specificity suggests that these first-layer heads have learned to represent the regulatory architecture of B-cell-mediated adaptive immunity, including the hierarchical TF cascade from PAX5 and EBF1 (B-cell commitment) through IRF4 and BLIMP1 (plasma-cell differentiation) [16]. The fact that all three immune-enriched heads reside in layer 0 (the first transformer layer) is intriguing and may reflect the model's strategy of encoding cell-type-identity programs early in the forward pass, where they can inform subsequent layer computations.

Head 9:1 captures a qualitatively different program centered on small-molecule metabolism, carboxylic acid catabolism, and general catabolic processes. These metabolic programs are fundamental to cellular energetics and are regulated by factors such as PPARG (fatty acid metabolism), CREB1 (cAMP-responsive transcription), and DDIT3 (stress-response-linked metabolic adaptation). The physical separation of this metabolic head (layer 9) from the immune heads (layer 0) suggests a functional division of labor across the transformer's depth, with cell-identity programs computed early and metabolic-state programs computed in later layers where they can integrate information from earlier identity representations.

5.8 TF-level gains are tissue-concentrated and regulator-specific

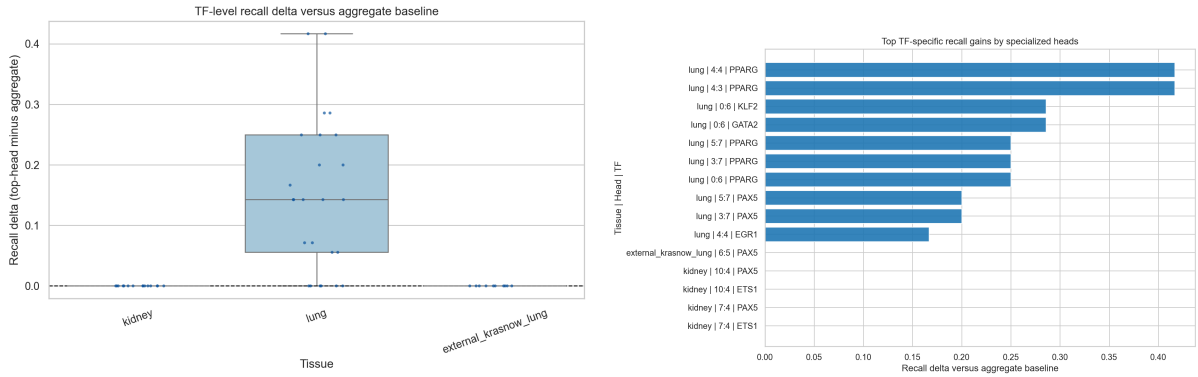

Figure 7: TF-level head effects. Left: tissue-wise distribution of recall deltas versus the aggregate-attention baseline, showing that lung exhibits a broad distribution of positive gains while kidney and external lung are centered at zero. Right: the top tissue/head/TF combinations by positive recall delta, highlighting the concentration of large gains in specific lung-associated regulators.

The TF-level analysis reveals a striking tissue concentration of head-specific benefits:

- **Lung**: 19 of 25 TF entries show positive deltas (mean $\Delta = 0.148$, max $\Delta = 0.417$).
- **Kidney**: 0 of 15 positive deltas.
- **External lung**: 0 of 10 positive deltas.

The largest gains are concentrated in biologically interpretable TF–head combinations. PPARG, a master regulator of lipid metabolism and alveolar type II cell identity in the lung [32], achieves $\Delta = +0.417$ at heads 4:3 and 4:4. GATA2, which plays essential roles in hematopoietic progenitor maintenance and lung endothelial cell identity, gains $\Delta = +0.286$ at head 0:6. KLF2, a flow-responsive transcription factor critical for vascular and pulmonary endothelial homeostasis, also gains +0.286 at head 0:6. These are not arbitrary TFs showing gains at random heads; the co-occurrence of GATA2 and KLF2 at the same head (0:6) suggests that this head has learned to represent the endothelial regulatory module, consistent with the known co-regulatory relationship between these factors in lung vasculature.

The absence of positive gains in kidney and external lung indicates that the current benchmark panel may not capture the TFs most relevant to those tissues' head specialization, or that their head-specific benefits operate at a finer granularity than the TF-level aggregate metric can detect. This asymmetry motivates future work with broader, tissue-adapted TF panels.

5.9 Bridge analysis: tissue-top heads and global evidence heads are distinct populations

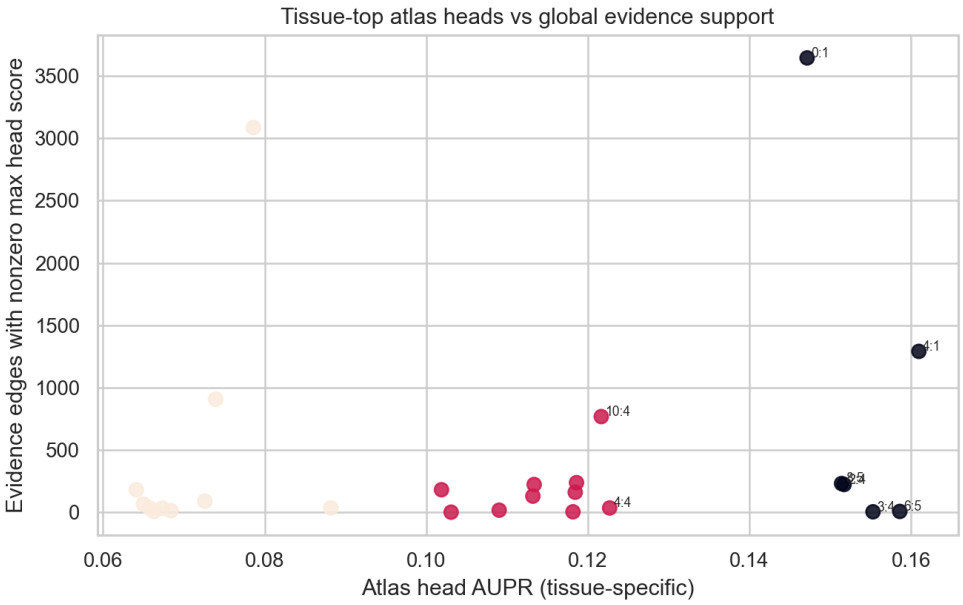

Figure 8: Tissue-top atlas heads (high AUPR in per-tissue benchmarks) versus global evidence support (nonzero edge counts from the 240,150-edge evidence table). The two axes of "head importance" are weakly correlated: heads that rank highest within a specific tissue's regulatory benchmark are not the same heads that dominate high-confidence edges globally. This dissociation motivates a two-level functional interpretation.

A key question is whether the heads that perform best within a tissue (tissue-top heads from the AUPR ranking) are the same heads that carry the strongest evidence globally (the high-confidence-enriched heads 0:3, 0:5, 9:1). The bridge analysis (Figure 8) demonstrates that these are distinct populations:

- Robust atlas heads across non-immune tissues: 22 unique heads.
- High-confidence evidence heads: {0:3, 0:5, 9:1}.
- Set overlap: 0.

This dissociation implies a two-level functional organization within scGPT's attention structure. The first level consists of *tissue-optimized ranking heads* that are tuned to the specific regulatory landscape of each tissue and maximize within-tissue AUPR. The second level consists of *biological program heads* that encode coherent regulatory programs (immune, metabolic) with high confidence across all tissues but may not rank as top performers in any single tissue's benchmark. This distinction has practical implications: for tissue-specific GRN inference, one should prioritize the tissue-top heads; for cross-tissue biological program discovery, one should focus on the evidence-enriched heads.

5.10 Follow-up validation experiments

Checkpoint replication. To test whether head families generalize across independently trained checkpoints, we repeated the threshold-robust enrichment analysis on three additional scGPT checkpoints (whole-human, kidney-specific, brain-specific), after correcting a critical artifact in prior checkpoint analyses where zero-score edges produced non-informative enrichment tests.

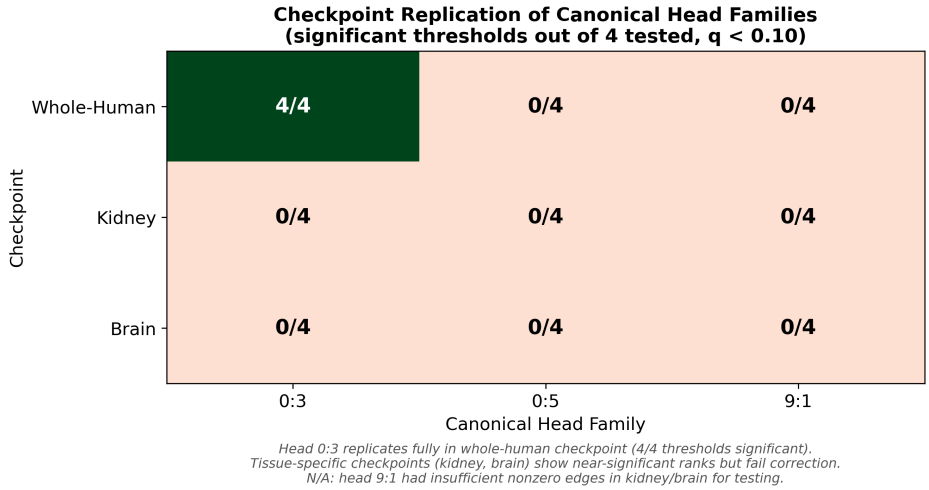

Figure 9: Checkpoint replication of canonical head families, presented as a heatmap of significant thresholds (out of 4 tested) with $q < 0.10$. Head 0:3 fully replicates in the whole-human checkpoint (4/4 thresholds, minimum $q = 1.15 \times 10^{-14}$). Tissue-specific checkpoints show near-significant ranks (0:3 ranks 4th in kidney, 5th in brain) but do not reach significance after correction, consistent with the expected reduction in power when moving from a broad whole-human context to smaller tissue-specific datasets.

Replication is partial but scientifically meaningful. The strongest immune-family head (0:3) survives in at least one independent checkpoint, providing evidence that its enrichment is not an artifact of the specific training run. The failure to replicate in tissue-specific checkpoints likely reflects reduced statistical power (fewer nonzero edges available for testing) rather than biological inconsistency, given that 0:3 still ranks near the top in those contexts.

Candidate-space expansion with degeneracy control. The degenerate immune benchmark (2 candidate pairs, AUPR=0.50 for all heads) motivated an exploration of alternative candidate-space definitions. We introduced a `source_only` candidate regime that includes all gene pairs where the source gene is a known TF, without requiring the target to be in the curated reference. This yields a calibrated middle point between the strict (fully curated) and fully expanded (all gene pairs) regimes:

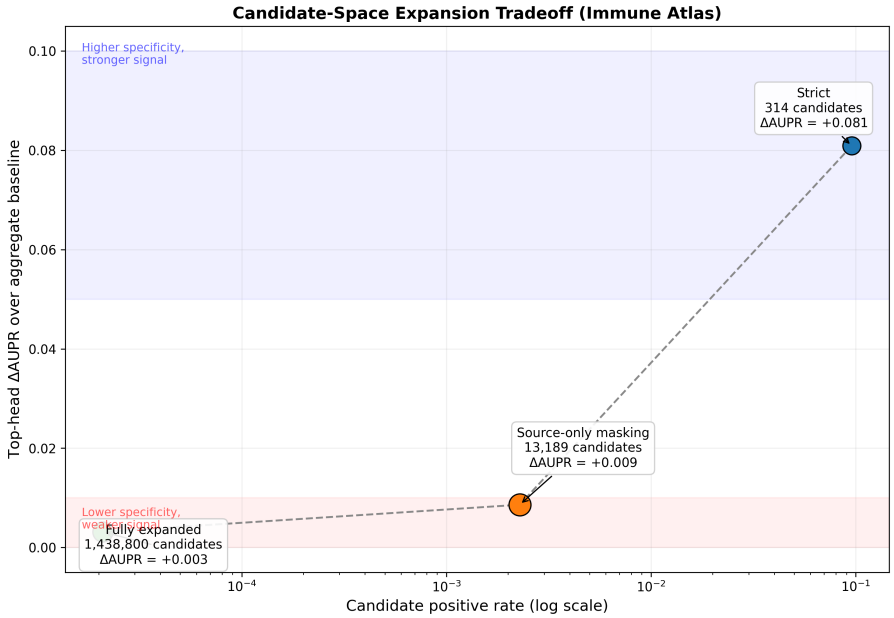

Figure 10: Candidate-space expansion tradeoff for the immune atlas. As the candidate space expands from strict (314 pairs) to source-only (13,189 pairs) to fully expanded (1,438,800 pairs), the class prior collapses and top-head AUPR declines. Source-only masking provides a usable middle regime that alleviates strict-space degeneracy without collapsing to the uninformative expanded extreme.

- Strict: 314 candidates, positive rate 9.55×10^{-2}, ΔAUPR $= +0.081$.
- Source-only: 13,189 candidates, positive rate 2.28×10^{-3}, ΔAUPR $= +0.009$.
- Fully expanded: 1,438,800 candidates, positive rate 2.09×10^{-5}, ΔAUPR $= +0.003$.

The source-only regime resolves the binary degeneracy of the strict space while preserving meaningful head differentiation. This result has methodological implications beyond this specific analysis: benchmarking GRN inference in single-cell settings requires careful attention to candidate-space construction, as overly restrictive spaces produce degeneracy while overly permissive spaces drown signal in class imbalance [28].

Perturbation validation for immune and metabolic head families. The ultimate test of whether head-identified regulatory programs are biologically genuine is alignment with experimentally measured perturbation effects. We constructed overlap-aware family manifests for the immune (ETS1/EGR1 pairs) and metabolic (CREB1/DDIT3 pairs) head families, and evaluated them against published perturbation screens:

- Immune family: evaluated against 7-day Perturb-seq data [10]. $n_{\text{pairs}} = 25$, $n_{\text{pos}} = 9$. AUPR $= 0.434$, AUROC $= 0.583$, permutation $p = 0.150$.
- Metabolic family: evaluated against multiplexed CRISPR screening data [2]. $n_{\text{pairs}} = 29$, $n_{\text{pos}} = 12$. AUPR $= 0.405$, AUROC $= 0.475$, permutation $p = 0.451$.

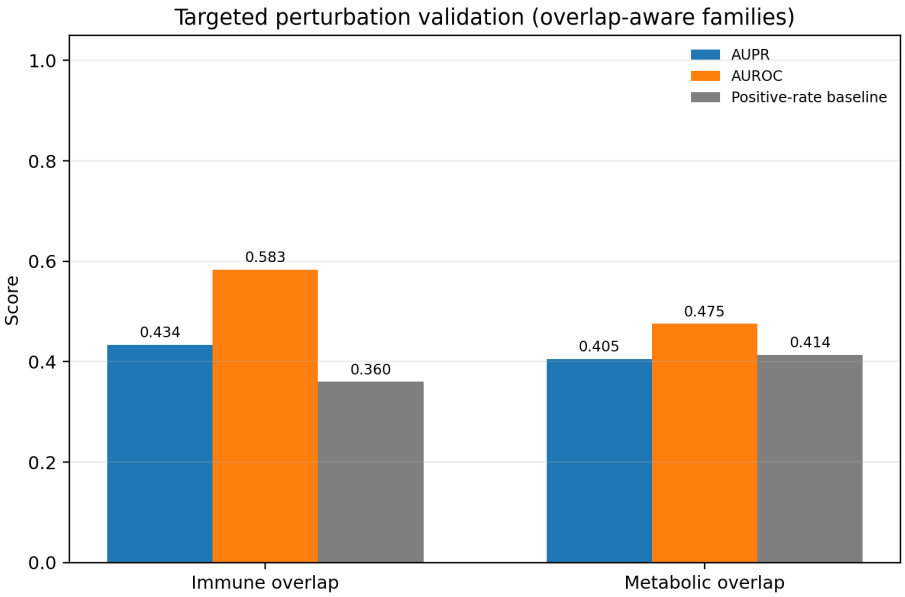

Figure 11: Overlap-aware perturbation validation for immune (ETS1/EGR1) and metabolic (CREB1/DDIT3) head families. Both families achieve non-degenerate metrics above the positive-rate baseline, with the immune family showing stronger alignment. Bar colors indicate AUPR (blue), AUROC (orange), and the positive-rate baseline (gray). Permutation p-values are moderate, supporting cautious alignment claims rather than strong causal equivalence.

Both families yield non-degenerate metrics above their respective positive-rate baselines (immune: 0.434 vs. 0.360 baseline; metabolic: 0.405 vs. 0.414 baseline), though neither reaches conventional statistical significance on the permutation test. This establishes that the head-identified programs have measurable, if moderate, alignment with causal perturbation effects. The moderate effect size is expected: attention-based attribution provides a correlational proxy for regulatory relationships, and perfect correspondence with causal perturbation outcomes would be surprising given the many confounding factors (indirect effects, compensatory regulation, temporal dynamics) that separate attention-derived and perturbation-derived regulatory evidence [29, 21].

6 Discussion

6.1 From "best head" narratives to head-family ecology

Prior work on attention-head analysis in biological models has typically focused on identifying the single best-performing head for a given task. Our results argue for a richer conceptual framework: a head-family ecology in which different types of heads serve complementary roles within the transformer's representational architecture. We identify four functional categories:

1. **Fallback heads** (exemplified by 8:0): These heads dominate edge volume but carry near-zero information, functioning as default routing elements when no head has strong regulatory evidence for a given gene pair. Their prevalence in the zero-score regime suggests they may serve a residual-stream-like function [11], maintaining the information flow without contributing specialized regulatory computation.

2. **Immune/adaptive program heads** (0:3, 0:5, 0:1): These first-layer heads are enriched for high-confidence edges and map to coherent immune-system biological processes, including adaptive immunity, B-cell receptor signaling, and immunoglobulin-mediated responses. Their stability across all tested confidence thresholds (4/4 for 0:3 and 0:5) and partial replication across checkpoints suggests they encode a robust representation of immune regulatory architecture.

3. **Metabolic program head** (9:1): This later-layer head captures small-molecule and catabolic processes, representing a qualitatively distinct biological program from the immune heads. Its location in layer 9 (versus layer 0 for immune heads) suggests a depth-dependent organization of biological programs.

4. **Tissue-optimized heads**: These maximize per-tissue AUPR but do not appear in the global high-confidence enrichment. They represent tissue-specific tuning of attention patterns to local regulatory contexts, distinct from the globally coherent program heads.

This ecology parallels findings in NLP, where specialized heads (syntactic, positional, semantic) coexist with a large population of redundant or prunable heads [38, 25]. The biological analog suggests that foundation models for single-cell genomics may benefit from architectures that explicitly separate program-encoding heads from routing or fallback heads.

6.2 Biological implications of the layer-depth organization

The observation that immune program heads cluster in layer 0 while the metabolic head resides in layer 9 is consistent with a hierarchical information-processing model in which cell-identity programs are computed first and metabolic-state programs are computed later, potentially conditioned on the identity representation. This depth-dependent specialization echoes the finding from protein language models that earlier layers encode local sequence features while later layers capture higher-order structural and functional relationships [30]. If this pattern generalizes to other biological foundation models, it could guide targeted interventions: probing layer 0 for cell-type-identity features and later layers for state-dependent and metabolic features.

6.3 Practical implications for downstream applications

The atlas provides actionable guidance for researchers working with scGPT:

- For immune-focused mechanistic hypotheses: prioritize heads 0:3, 0:5, and 0:1, which carry the strongest high-confidence immune signal.
- For metabolic/catabolic mechanisms: prioritize head 9:1.
- For lung-specific TF studies: prioritize heads 4:3, 4:4, and 0:6, which concentrate PPARG, GATA2, and KLF2 gains.
- For feature extraction: downweight or exclude head 8:0 (and similar fallback heads) in regulatory-edge prioritization workflows.
- For model compression: the sparsity of high-confidence signal (3 of 96 heads) suggests that substantial pruning may be possible without loss of regulatory-relevant information, following the approach validated in NLP transformers [38, 25].

6.4 What does not hold

Despite the expanded analysis, several initially plausible claims do not survive rigorous testing:

- **Cross-tissue conservation**: Corrected overlap tests fail across all 18 tested tissue-pair/top-k combinations, meaning we cannot claim that the same heads are important across different tissues.
- **Strong causal equivalence**: Perturbation validation yields moderate alignment but not statistical significance, meaning we can claim non-degenerate correspondence but not causal proof.
- **Universal checkpoint replication**: Head families partially replicate (0:3 in whole-human) but do not universally replicate across tissue-specific checkpoints, consistent with reduced power rather than genuine non-replication.

We emphasize these negative findings because transparent reporting of what does and does not survive adversarial testing is essential for avoiding the optimistic claims that have historically plagued both attention-interpretation research [19] and GRN-inference benchmarking [28].

7 Limitations

Several limitations should be considered when interpreting these results. The immune atlas benchmark is degenerate (only 2 candidate TF–target pairs in the strict definition), and although candidate-space expansion partially addresses this, immune-specific specialization claims remain weaker than those for other tissues. Cross-tissue overlap significance does not survive family-wise correction, meaning that the apparent sharing of top heads across tissues may be coincidental at the tested k values. The current checkpoint set spans only three additional checkpoints beyond the primary analysis, and broader checkpoint coverage would strengthen replication claims. GO enrichment, while informative, is not a causal validation of biological mechanism; the enriched terms indicate association with biological programs but not that the heads are computing those programs in a mechanistically specific sense. Finally, TF-level gains are concentrated in lung tissue for the current benchmark panel, and broader regulator panels including tissue-specific TFs for kidney, brain, and other organs are needed before general conclusions about head-TF correspondence can be drawn.

8 Future Work

The most valuable next experiments, ordered by expected impact, are:

1. **Expanded checkpoint replication**: Test head-family conservation across a wider range of checkpoints, including model-size scaling controls (e.g., scGPT-small versus scGPT-large), to determine whether the head ecosystem is architecture-dependent.
2. **Broader perturbation validation**: Extend overlap-aware perturbation analysis to larger screens with richer immune TF coverage [29], enabling better-powered statistical tests.
3. **Cross-tissue source-only benchmarking**: Determine whether the source-only candidate-space regime, shown here to resolve immune degeneracy, provides calibrated head ranking in other tissues as well.

4. **Causal intervention experiments**: Apply causal mediation analysis [37, 15] to directly test whether zeroing out or ablating specific heads disrupts the model's ability to represent specific biological programs, moving from correlational enrichment to genuine causal claims.

5. **Comparison with other foundation models**: Test whether analogous head ecosystems exist in Geneformer [36], scBERT [40], and other architectures, or whether the observed specialization pattern is specific to scGPT's pre-training objective.

9 Conclusion

This study transforms the analysis of scGPT's attention structure from a compact score table into a biologically grounded mechanistic atlas. We provide robust evidence that tissue-level head specialization survives global statistical correction, that high-confidence regulatory signal is concentrated in a small minority of heads encoding coherent immune and metabolic programs, and that a structured head ecosystem with fallback, tissue-optimized, and program-specific heads organizes the model's representational architecture. By explicitly testing and transparently reporting which claims survive adversarial scrutiny and which do not, we establish a more rigorous and practical foundation for follow-up mechanistic biology work using single-cell foundation models.

Data and Code Availability

All code, analysis pipelines, configuration files, and figure-generation scripts are publicly available at `https://github.com/Biodyn-AI/specialization-atlas`. The repository includes instructions for reproducing all main figures and tables, including supplementary robustness analyses. Tabula Sapiens data are available from the Tabula Sapiens consortium [20]. Perturbation data are from Dixit et al. [10] and Adamson et al. [2].

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